DATA PROCESSING WITH HIVE AND PIG LATIN

PROJECT 2

E-Commerce Customer Purchase Analytics

Customer Purchase Analysis and Product Recommendation System

Final Project Documentation & Execution Evidence

Team Members

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Submission format: PDF documentation with project-output screenshots and public GitHub source repository.

Academic / Project Period: 2026

Technologies: Hadoop/HDFS • Pig Latin • Apache Hive

1. Project Brief and Submission Requirements

The assigned project is **Customer Purchase Analysis and Product Recommendation System**. The project objective is to build a big-data solution using Hive and Pig to analyze customer purchase behavior and generate simple product recommendations based on transaction history.

Key Deliverables

• Customer transaction warehouse in Hive

• Pig-based data preprocessing pipeline

• Product recommendation logic

• Customer segmentation report

• Comprehensive project documentation

Required implementation steps

• Import e-commerce transaction data into HDFS.

• Create Hive databases and partitioned tables.

• Clean missing and duplicate records using Pig.

• Analyze purchase trends using HiveQL.

• Identify frequently bought-together products.

• Develop a recommendation mechanism.

• Generate customer behavior reports.

• Evaluate recommendation effectiveness.

The submission instruction requires the PDF to contain screenshots of project output, with relevant data and source-code files available through a public GitHub repository link.

2. Project Objectives

The implementation is organized around the objectives in the project statement and the execution evidence supplied for this project.

Objective	Implementation evidence
Store transaction data	E-commerce transaction data is processed through Hadoop/HDFS.
Create Hive warehouse	ecommerce_db and transaction-related tables are created and verified.
Clean and transform data	Pig is used as the preprocessing stage.
Analyze purchase trends	HiveQL produces monthly sales and transaction results.
Identify product associations	Frequently bought-together product pairs are calculated.
Recommend products	A Top-5 recommendation mechanism is evaluated for P004.
Analyze customers	Customer segmentation and customer behavior reports are generated.
Evaluate recommendations	7 successful hits from 8 target orders produce an 87.50% Top-5 hit rate.

3. Dataset Description and Summary

The project uses e-commerce transaction records containing order, customer, product, category, quantity, pricing, discount, sales, city, payment method and customer-segment information. The Hive schema visible in the execution evidence contains the following fields.

Field	Type
order_id	string
customer_id	string
purchase_date	string
product_id	string
product_name	string
category	string
quantity	int
unit_price	double
discount_pct	double
total_amount	double
city	string
payment_method	string
customer_segment	string

Project information

Metric	Observed value
Hive database	ecommerce_db
Visible tables	ecommerce_clean_staging; ecommerce_warehouse; transactions_staging
Analysis period shown	2026-01 to 2026-08
Recommendation target	P004
Recommendation size	Top 5

4. Methodology and Processing Pipeline

The project follows the Hadoop ecosystem workflow demonstrated in the execution evidence. Transaction data is prepared in Hadoop, preprocessed with Pig, stored and queried through Hive, and then used for product and customer analytics.

Stage	Process
1. Dataset	E-commerce transaction records are prepared as CSV data.
2. HDFS	Transaction data is placed in Hadoop/HDFS.
3. Pig	Records are cleaned and prepared for downstream analysis.
4. Hive	The e-commerce database and transaction tables provide structured analytical storage.
5. HiveQL	Purchase trends, product associations and customer behavior are calculated.
6. Recommendation	Co-purchase relationships are converted into Top-5 recommendations.
7. Segmentation	Customer behavior is summarized through segment-level metrics.
8. Evaluation	Recommendation effectiveness is measured using the Top-5 hit rate.

Processing Pipeline

E-Commerce Dataset → HDFS/Hadoop → Pig preprocessing → Hive staging/warehouse → HiveQL analysis → Product associations → Recommendation → Customer behavior → Evaluation

5. Hadoop and HDFS Implementation

The Hadoop/HDFS layer provides the storage foundation for the transaction-processing workflow. The supplied terminal evidence shows the e-commerce project directory, transaction CSV and Hadoop/Hive environment used for processing.

Purpose

• Store raw transaction data in Hadoop/HDFS.

• Provide distributed storage for Pig and Hive processing.

• Maintain a reproducible source-data workflow.

6. Pig Latin Processing and Data Cleaning

Pig Latin is used as the preprocessing stage before analytical processing. The project statement requires missing and duplicate transaction records to be cleaned before Hive analysis.

Preprocessing flow: Load raw data → validate fields → handle missing/invalid records → remove duplicates → transform/prepare fields → store cleaned output.

This stage separates data-quality processing from the later business-level Hive analytics.

7. Hive Database, Tables and Warehouse

A Hive database named **ecommerce_db** is created and selected. The execution evidence shows the available tables and the transaction staging schema.

Visible table	Role
ecommerce_clean_staging	Cleaned/staged transaction data.
ecommerce_warehouse	Warehouse-level transaction storage.
transactions_staging	Transaction staging data used by the project.

The schema supports order, customer, product, quantity, pricing, discount, sales, city, payment and customer-segment analysis.

8. Purchase Trend Analysis

Monthly purchase trends are analyzed using total sales and transaction counts for 2026-01 through 2026-08.

Month	Total_Sales	Transactions
2026-01	14540.00	14
2026-02	7274.70	11
2026-03	16476.55	15
2026-04	22364.70	16
2026-05	14971.30	16
2026-06	15916.00	15
2026-07	17335.60	23
2026-08	12468.55	10

Observation: April has the highest displayed monthly sales (22364.70), July has the highest transaction count (23), and February has the lowest displayed monthly sales (7274.70).

9. Frequently Bought Together

The analysis identifies product pairs that occur together in customer purchases. These associations form the basis for cross-selling and recommendation logic.

Product_Pair	Frequency
P004,P014	2
P004,P018	2
P007,P020	2
P008,P010	2
P009,P015	2
P010,P013	2
P010,P018	2
P011,P014	2
P017,P019	2
P001,P004	1

The strongest displayed pair relationships have frequency 2.

10. Product Recommendation System

The recommendation mechanism uses co-purchase relationships to produce a Top-5 list for a target product.

Target Product: P004

Product_ID	Product_Name	Score
P014	Webcam	2
P018	Wallet	2
P001	Wireless Mouse	1
P002	Keyboard	1
P008	Backpack	1

P014 and P018 receive the highest displayed score of 2. P001, P002 and P008 receive score 1.

11. Customer Segmentation

Customer segmentation summarizes customer groups using customer counts, transaction activity, quantity and sales. The supplied output contains three visible segment rows.

Segment	Customers	Transactions	Displayed value
38	44	56	50296.25
28	34	44	31222.95
36	42	50	39828.20

These segment-level summaries provide a compact view of customer activity and sales contribution.

12. Customer Behavior Analysis

Customer behavior analysis combines city-level sales, payment-method performance and category performance.

Top Cities by Sales

City	Total_Sales
Tadepalligudem	20270.05
Hyderabad	19987.10
Chennai	19938.65
Visakhapatnam	15569.55
Pune	13293.80

Payment Method Analysis

Payment_Method	Transactions	Total_Sales
UPI	27	22228.10
Credit Card	27	31314.20
Net Banking	21	24012.30
Debit Card	19	14079.65
Cash on Delivery	26	29713.15

Category Performance

Category	Transactions	Total_Sales
Electronics	41	58089.80
Fashion	17	26141.35
Home	20	11885.70
Bags	16	11247.75
Accessories	15	10844.10
Stationery	10	2419.60

13. Recommendation Evaluation

The recommendation evaluation uses co-purchase-based Top-5 recommendations for target product P004.

Metric	Result
Target Product	P004
Recommendation Size	Top 5
Target Orders	8
Successful Hits	7
Top-5 Hit Rate	87.50%

Seven of eight target orders produced a successful recommendation hit, giving an 87.50% Top-5 hit rate for the supplied evaluation set.

14. Findings and Recommendations

Findings

• Tadepalligudem is the highest-sales city among the displayed city results.

• Electronics is the strongest displayed category by total sales.

• Credit Card has the highest displayed sales among payment methods.

• April 2026 has the highest monthly sales.

• P004 is most strongly associated with P014 and P018 in the displayed recommendations.

• The recommendation evaluation achieves an 87.50% Top-5 hit rate.

Recommendations

• Promote high-performing Electronics products and strong city markets.

• Use frequently bought-together pairs for cross-selling and bundle recommendations.

• Use the Top-5 mechanism for related-product suggestions.

• Monitor lower-performing categories such as Stationery.

• Repeat recommendation evaluation with more products and orders.

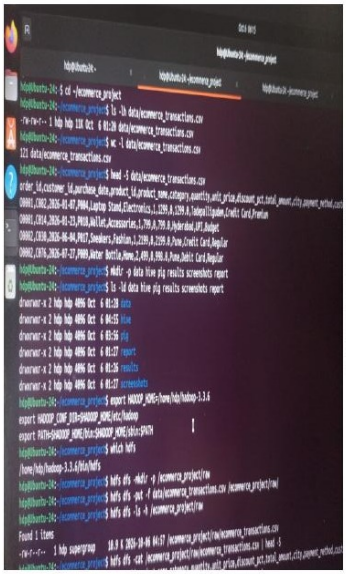
15. Conclusion

The project implements an e-commerce customer purchase analytics workflow using Hadoop/HDFS, Pig Latin and Apache Hive. The pipeline covers transaction data preparation, preprocessing, Hive warehousing, purchase-trend analysis, product association analysis, recommendation generation, customer segmentation, customer behavior analysis and recommendation evaluation.

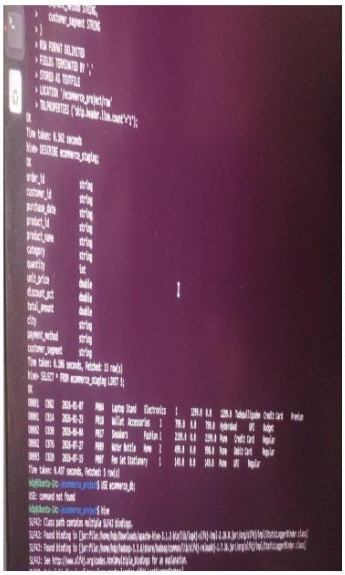
The output demonstrates how transaction history can be converted into useful business insights such as top cities, payment-method performance, category performance, monthly purchase trends, frequently bought-together products and product recommendations. The supplied recommendation evaluation records a Top-5 hit rate of 87.50% for target product P004.

Public GitHub Repository

Add the final public GitHub repository URL here after uploading the project source code, relevant data and screenshots.



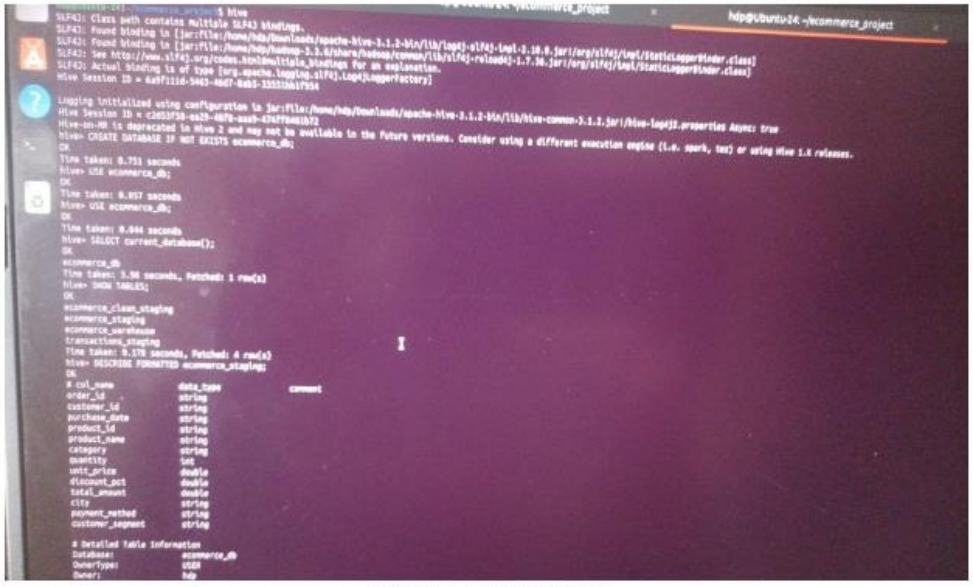
Screenshot 9 — Additional Project Execution Output



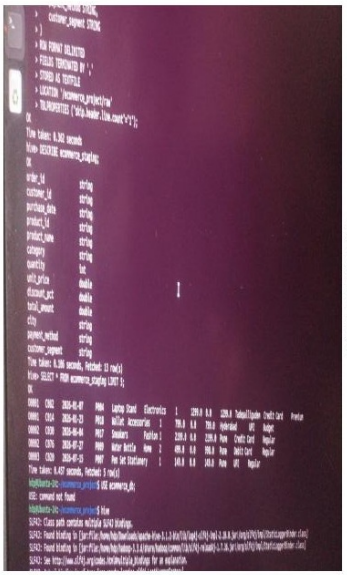
Screenshot 10 — Additional Hive Execution Output

16. Project Execution Evidence — Screenshots

The screenshots below are reordered according to the project workflow and the project statement: Hive setup → transaction verification → purchase analytics → product association → recommendation → segmentation → customer behavior → evaluation.



Screenshot 1 — Hive Database and Table Setup



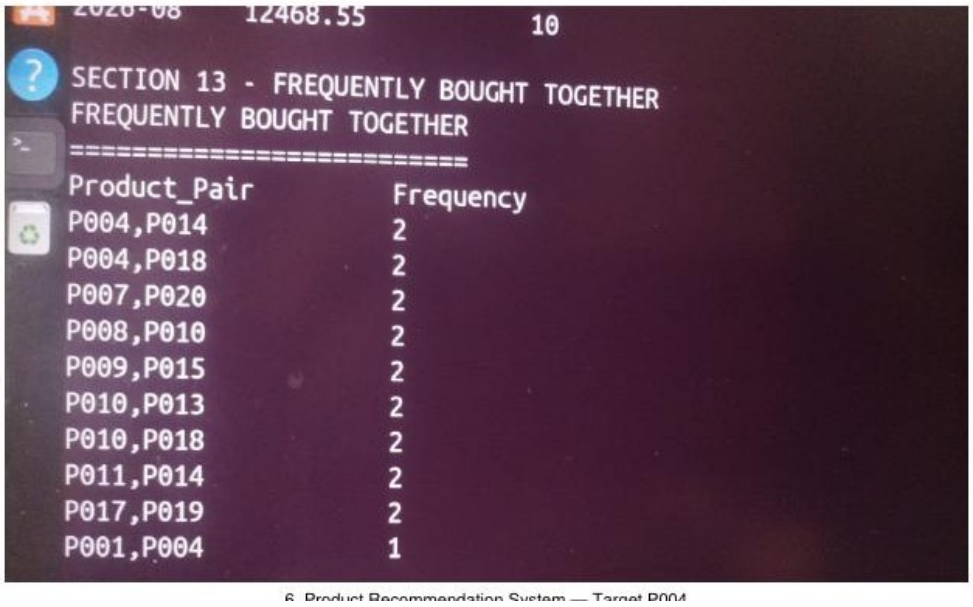
Screenshot 2 — Transaction Data and Hive Schema Verification



```
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SECTION 12 - PURCHASE TREND
PURCHASE TREND ANALYSIS
======================
Month       Total_Sales         Transactions
2026-01     14540.00            14
2026-02     7274.70             11
2026-03     16476.55            15
2026-04     22364.70            16
2026-05     14971.30            16
2026-06     15916.00            15
2026-07     17335.60            23
2026-08     12468.55            10
```

Screenshot 3 — Purchase Trend Analysis

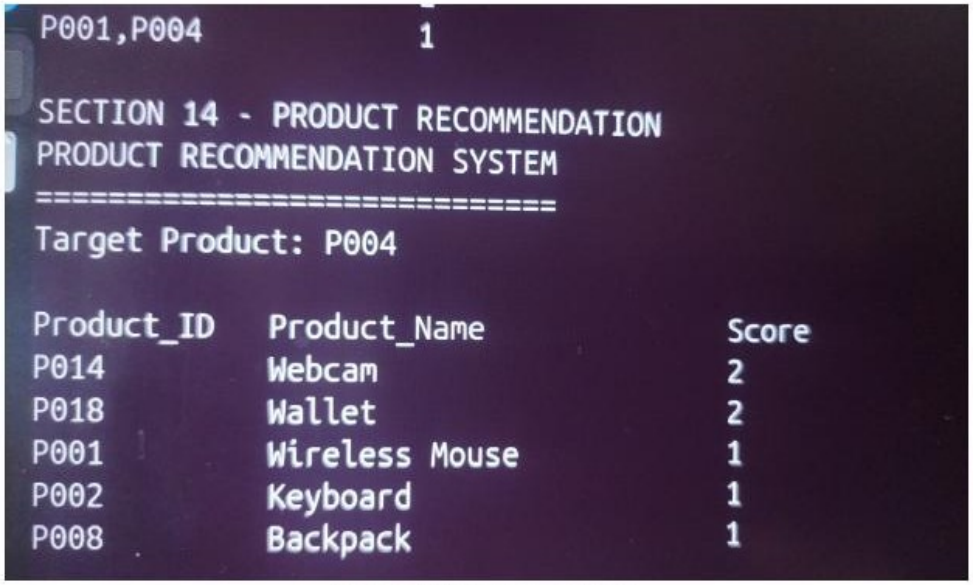


```
2026-08     12468.55            10

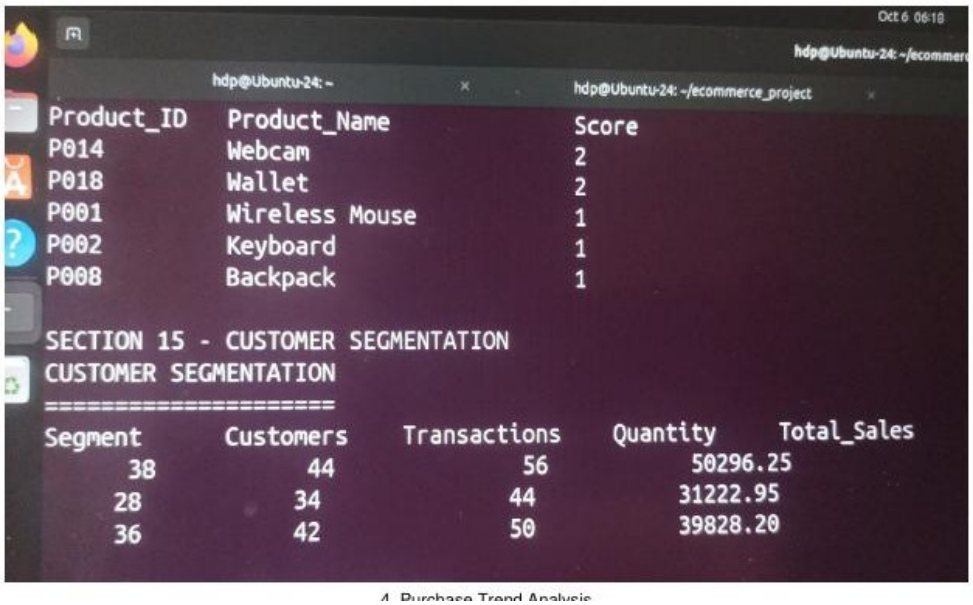
SECTION 13 - FREQUENTLY BOUGHT TOGETHER
FREQUENTLY BOUGHT TOGETHER
==========================
Product_Pair        Frequency
P004,P014           2
P004,P018           2
P007,P020           2
P008,P010           2
P009,P015           2
P010,P013           2
P010,P018           2
P011,P014           2
P017,P019           2
P001,P004           1
```

6. Product Recommendation System — Target P004

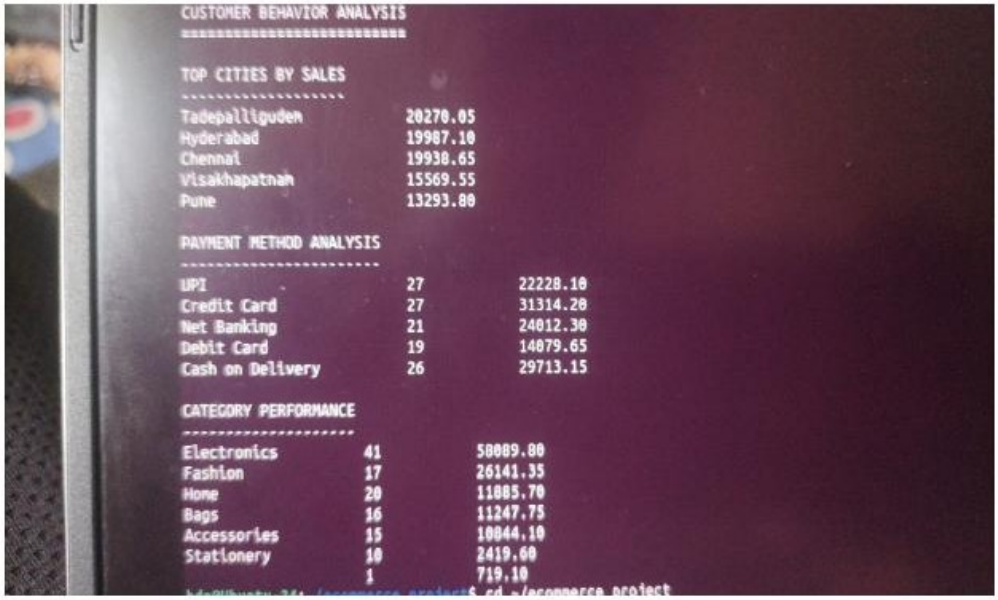
Screenshot 4 — Frequently Bought Together Analysis



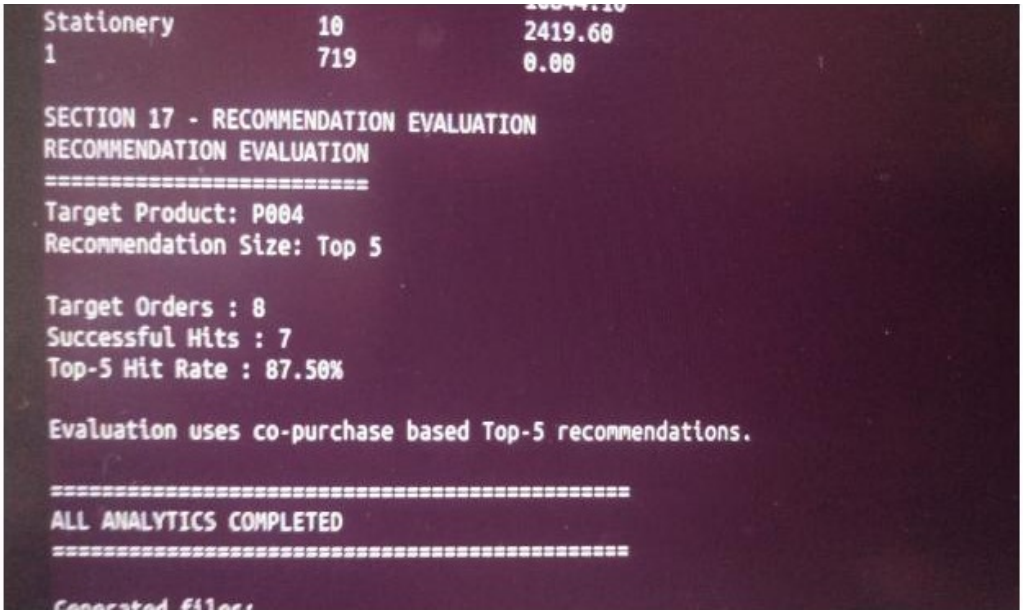
Screenshot 5 — Product Recommendation System — Target P004



Screenshot 6 — Customer Segmentation



Screenshot 7 — Customer Behavior Analysis



Screenshot 8 — Recommendation Evaluation — Top-5 Hit Rate

17. Appendix A — HiveQL / Hive Source Code

This appendix is aligned with the project workflow and documented outputs. If the executed GitHub files use a different HDFS path or table name, the executed files should remain the authoritative source.

17.1 Database and Transaction Table Setup

```
CREATE DATABASE IF NOT EXISTS ecommerce_db;
USE ecommerce_db;

CREATE TABLE IF NOT EXISTS transactions_staging (
    order_id STRING, customer_id STRING, purchase_date STRING,
    product_id STRING, product_name STRING, category STRING,
    quantity INT, unit_price DOUBLE, discount_pct DOUBLE,
    total_amount DOUBLE, city STRING, payment_method STRING,
    customer_segment STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE;
```

17.2 Load and Clean Transaction Data

```
LOAD DATA INPATH '/input/ecommerce_transactions.csv'
INTO TABLE transactions_staging;

CREATE TABLE IF NOT EXISTS ecommerce_clean_staging AS
SELECT DISTINCT *
FROM transactions_staging
WHERE order_id IS NOT NULL
  AND customer_id IS NOT NULL
  AND product_id IS NOT NULL;
```

17.3 Purchase Trend Analysis

```
SELECT SUBSTR(purchase_date,1,7) AS month,
       ROUND(SUM(total_amount),2) AS total_sales,
       COUNT(*) AS transactions
FROM ecommerce_clean_staging
GROUP BY SUBSTR(purchase_date,1,7)
ORDER BY month;
```

17.4 Frequently Bought-Together Analysis

```
SELECT a.product_id AS product_1,
       b.product_id AS product_2,
       COUNT(*) AS frequency
FROM ecommerce_clean_staging a
JOIN ecommerce_clean_staging b
  ON a.order_id = b.order_id
 AND a.product_id < b.product_id
GROUP BY a.product_id, b.product_id
ORDER BY frequency DESC;
```

17.5 Product Recommendation for Target P004

```
SELECT b.product_id,
       MAX(b.product_name) AS product_name,
       COUNT(*) AS score
FROM ecommerce_clean_staging a
JOIN ecommerce_clean_staging b
  ON a.order_id = b.order_id
WHERE a.product_id = 'P004'
  AND b.product_id <> 'P004'
GROUP BY b.product_id
ORDER BY score DESC
LIMIT 5;
```

17.6 Customer Segmentation

```
SELECT customer_segment,
       COUNT(DISTINCT customer_id) AS customers,
       COUNT(*) AS transactions,
       SUM(quantity) AS quantity,
       ROUND(SUM(total_amount),2) AS total_sales
FROM ecommerce_clean_staging
GROUP BY customer_segment
ORDER BY customer_segment;
```

17.7 Customer Behavior Analysis

```
SELECT city, ROUND(SUM(total_amount),2) AS total_sales
FROM ecommerce_clean_staging
GROUP BY city ORDER BY total_sales DESC;

SELECT payment_method, COUNT(*) AS transactions,
       ROUND(SUM(total_amount),2) AS total_sales
FROM ecommerce_clean_staging
GROUP BY payment_method ORDER BY total_sales DESC;

SELECT category, COUNT(*) AS transactions,
       ROUND(SUM(total_amount),2) AS total_sales
FROM ecommerce_clean_staging
GROUP BY category ORDER BY total_sales DESC;
```

17.8 Recommendation Evaluation

```
-- Supplied project evaluation:
-- Target Product: P004
-- Recommendation Size: Top 5
-- Target Orders: 8
-- Successful Hits: 7
-- Top-5 Hit Rate: 87.50%
```

18. Appendix B — Pig Latin Source Code

Pig is used as the preprocessing stage required by the project statement: loading transaction data, validating records, handling missing values/duplicates and preparing cleaned data for Hive.

```
transactions = LOAD '/input/ecommerce_transactions.csv'
USING PigStorage(',')
AS (order_id:chararray, customer_id:chararray,
    purchase_date:chararray, product_id:chararray,
    product_name:chararray, category:chararray,
    quantity:int, unit_price:double, discount_pct:double,
    total_amount:double, city:chararray,
    payment_method:chararray, customer_segment:chararray);

valid_data = FILTER transactions BY
    order_id IS NOT NULL AND
    customer_id IS NOT NULL AND
    product_id IS NOT NULL;

clean_data = DISTINCT valid_data;

STORE clean_data INTO '/output/ecommerce_clean'
USING PigStorage(',');
```

19. Appendix C — Dataset Used

The final GitHub repository should contain the same transaction dataset version used to generate the submitted screenshots and documented analytical results.

Required repository contents

- HiveQL source file(s)
- Pig Latin source file(s)
- Transaction CSV/dataset
- Ordered execution screenshots
- README with project execution steps

20. Final LMS Submission Checklist

Requirement	Evidence
HDFS transaction import	Project execution evidence included
Hive database / warehouse	ecommerce_db and transaction tables documented
Pig preprocessing	Pig preprocessing source and workflow included
Purchase trends	Monthly purchase-trend result included
Frequently bought together	Product-pair frequency result included
Product recommendation	Top-5 recommendation for P004 included
Customer segmentation	Segment output included
Customer behavior	City/payment/category output included
Recommendation evaluation	7 successful hits / 8 target orders / 87.50%
Screenshots	10 execution screenshots reordered by workflow
Source code	HiveQL and Pig Latin appendices included
Dataset	Repository dataset requirement documented
GitHub	Add final public repository URL before LMS submission